

YASHASWINI KULSHRESTHA

Software Engineer (Backend + Agentic AI) | Spring Boot | LangGraph | LLM Systems | Final Year B.Tech IT, NSUT Delhi
+91-7701892117 kulshresthayashaswini@gmail.com LinkedIn GitHub LeetCode

Experience

Software Engineering Intern (Backend & Data Pipelines)

Jan 2026 – Present

HealthKart

Gurgaon

- Owned backend feature releases across **Spring Boot** services; designed a multi-layer caching architecture (**Redis** cache-aside + response caching + TTL invalidation) across product catalogue and user-facing APIs, cutting average API latency by **~40%** and eliminating redundant DB calls.
- Architected a config-driven analytics platform using **ClickHouse** and **CubeJS**, enabling dashboards to be created through JSON configurations with 20+ reusable components, eliminating custom frontend and backend development.
- Engineered a multi-tenant customer event-tracking system in **Python** using **Dagster** and **Segment CDP**, with automated asset checks for schema drift and null-rate anomalies to keep faulty data out of downstream production systems.

Software Engineering Intern

May 2025 – Jun 2025

Microsoft

Hyderabad

- Developed a **Python** static analysis tool scanning Windows C/C++ codebases for unsafe API patterns (**SetProcessMitigationPolicy**, DACL-affecting calls), reducing manual security review effort by **35%** and false negatives by **20%**.
- Integrated the tool into **CI/CD pipelines** and surfaced findings via a **Streamlit** dashboard for MSRC triage; parsed manifest files to correlate privilege context with vulnerable API combinations.

Education

Netaji Subhas University of Technology, Delhi

Aug 2022 – May 2026

B.Tech in Information Technology

CGPA: 8.3 / 10

Projects

ATS Resume Agent | Python, LangGraph, FastAPI, GPT-4o-mini, React, Docker, pdflatex | GitHub

2025

- Built a **full-stack agentic AI application** — FastAPI backend orchestrating a 7-node **LangGraph** pipeline (JD parsing → ATS scoring → gap analysis → LaTeX rewrite → PDF compile) with a React frontend; fully containerised via **Docker**.
- Enforced structured JSON output at every node via **Pydantic** while calling **GPT-4o-mini**; final node recompiles a real **pdflatex** PDF preserving the original template — end-to-end server-side document generation.

VeriLoan – Agentic Loan Onboarding System | GPT-4o, Whisper STT, Node.js, Express, MediaPipe, React | GitHub

2025

- Designed an agentic pipeline where **GPT-4o** performs live structured data extraction from Whisper-transcribed audio, but runs strictly in **shadow mode** — a deterministic rule engine (200+ rules) holds final decision authority, ensuring **RBI V-CIP** auditability.
- Implemented fraud detection covering 4 real-time scenarios (location mismatch, replay attack, income inconsistency) with rule-trace logging; added production hardening — rate limiting, Helmet.js headers, and immediate sensitive-data cleanup.

PersonalMail-RL | Python, TRL GRPO, Qwen2.5-1.5B, QLoRA, FastAPI, HuggingFace, Docker | GitHub

2026

- Finalist, Meta OpenEnv Hackathon Apr 2026.** Fine-tuned **Qwen2.5-1.5B-Instruct** via TRL GRPO with 4-bit QLoRA in a 2-step RL environment (classify email → draft reply), achieving **+37.9%** reply quality and **+42.6%** tone accuracy across 25 evaluation scenarios.
- Devised 7 verifiable reward functions with anti-reward-hacking safeguards; identified and patched a live type-unsafe **JSON output exploit** during GRPO training; deployed via FastAPI + Docker on HuggingFace Spaces (A100).

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, C++, Bash

Backend & APIs: Spring Boot, FastAPI, Node.js, Express, REST APIs, Redis Caching, OOP, Design Patterns

Agentic AI & LLM: LangGraph, LangChain, GPT-4o, Pydantic, Prompt Engineering, RAG, Multi-Agent Systems, LLM

Fine-Tuning (GRPO, QLoRA), MCP, LangSmith

Data & Pipelines: PostgreSQL, MySQL, ClickHouse, CubeJS, Dagster, Segment CDP, Schema Validation, Query Optimisation

Infra & Tools: Docker, CI/CD, AWS, Azure, Git, Streamlit, Tableau

Achievements

- Finalist, Meta OpenEnv Hackathon Apr 2026 (LLM fine-tuning track).
- Top 47 of 100,000+ participants, EY Hackathon 5.0 (team lead).
- Reliance Foundation Scholarship (INR 2,00,000) for academic excellence.
- Solved 700+ problems on LeetCode, GeeksforGeeks, and Codeforces.

Positions of Responsibility

Placement Coordinator

Aug 2024 – May 2026

Training & Placement Cell, NSUT Delhi

- Coordinated on-campus recruitment drives for **40+ companies** including Fortune 500 firms; served as primary liaison between recruiters and the student body, managing interview schedules, JD dissemination, and offer communication for **900+ students**.